PINO: Person-Interaction Noise Optimization for Long-Duration and Customizable Motion Generation of Arbitrary-Sized Groups

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Abstract

*Generating realistic multi-person interactions at scale remains an open challenge in human motion synthesis. Existing methods either rely on shared text prompts that yield uniform, undifferentiated behaviors, or require retraining when new constraints are introduced. We present PINO (**P**erson-**I**nteraction **N**oise **O**ptimization), a training-free framework that composes arbitrary-sized group interactions from a pretrained two-person diffusion model (InterGen). Our key insight is to decompose group dynamics into a sequence of pairwise generations anchored by a shared "pivot" character. For each new person added to the group, PINO conditions the target character on the pivot via masked diffusion and optimizes the initial noise with differentiable physics-based penalties, including overlap avoidance, root position, movement region, facing orientation, and relative positioning. Per-pair text prompts enable fine-grained role assignment without shared-prompt ambiguity. For long-duration sequences, PINO employs motion inpainting with boundary smoothness penalties to extend interactions while preserving temporal coherence. Experiments on the InterHuman dataset demonstrate that PINO eliminates body overlap (Overlap* ≈ 0.000 *vs.* 0.119 *for InterGen in the two-person case), dramatically reduces penetration volume (*275.65 *vs.* 3112.72*), and maintains competitive semantic quality (FID* 13.163 *vs.* 13.278*). These benefits scale to groups of five or more persons without any additional training.*

1. Introduction

Synthesizing realistic human motion has broad applications in computer animation, video games, and robotics. While single-person text-to-motion generation has achieved impressive results [1, 3, 20, 26], extending these techniques to multi-person interactions poses unique challenges: motions must be jointly plausible, semantically consistent with natural language descriptions, and physically coherent (i.e., free of body penetration and interpenetration artifacts).

Existing approaches to multi-person generation fall into three broad categories. First, methods based on ControlNet [25] (e.g., InterControl [22], FreeMotion [2]) adapt single-person or two-person networks through conditional guidance. Second, directly trained multi-person networks (e.g., Shan et al. [16]) are learned on multi-person datasets, limiting scalability. Third, diffusion-based composition methods such as PriorMDM [15] combine single-person priors via learned communication blocks. Across all categories, a common limitation is that adding new characters or introducing new spatio-temporal constraints typically requires either retraining or significant model redesign.

Inspired by Diffusion Noise Optimization (DNO) [7] and ProgMoGen [11], which demonstrated that optimizing the initial noise of a diffusion model—rather than its weights—enables zero-shot controllability for single-person motion, we extend this principle to the multi-person setting. Our approach, PINO (**P**erson-**I**nteraction **N**oise **O**ptimization), leverages a pretrained two-person diffusion model (InterGen [10]) and composes arbitrarily large groups through sequential, pairwise generation guided by physics-aware noise optimization.

The core idea of PINO is to treat group interactions as a chain of pairwise relationships anchored by a pivot character. When generating person k ($k > 2$), we fix the motion of the pivot (person 1) via feature-space masking and generate person k conditioned on the pivot using a per-pair text prompt. The initial noise for person k is optimized through a differentiable loss that combines: (i) an *overlap penalty* enforcing minimum distance between all character pairs; (ii) a *root position penalty* controlling start/end positions; (iii) a *movement region penalty* constraining spatial extents; (iv) a *facing orientation penalty* specifying relative body orientations; and (v) a *relative position penalty* governing pairwise spatial relationships. For long-duration generation, motion inpainting with a boundary smoothness penalty enables coherent sequence extension without retraining.

Our contributions are as follows:

- A **training-free** framework that scales a pretrained two-person diffusion model to arbitrary group sizes via sequential pairwise generation with pivot conditioning.
- **Physics-aware noise optimization** that enforces overlap avoidance and fine-grained spatio-temporal controls entirely at inference time.

- **Per-pair prompt switching** that enables nuanced role assignment and customizable behaviors within large groups, avoiding shared-prompt ambiguity.
- **Long-duration motion generation** via inpainting with smooth boundary transitions, enabling evolving interactions over time.

Experiments on the InterHuman benchmark show that PINO achieves near-zero overlap even in five-person scenarios, dramatic reductions in penetration volume, and competitive or superior FID scores compared to InterGen and FreeMotion, all without additional training.

2. Related Work

Single-Person Motion Generation. Text-conditioned human motion synthesis has made rapid progress using variational autoencoders [12, 13], transformer-based architectures [3], and diffusion models [1, 20, 26]. MDM [20] introduced a classifier-free diffusion model for motion that predicts the clean sample at each denoising step. MLD [1] performs diffusion in a learned latent space to reduce computational cost. These works focus on single-person scenarios and serve as the backbone components for multi-person extensions.

Controllable Motion Generation. Adding spatial constraints to diffusion models without retraining is an active research direction. GMD [6] introduces imputation-based spatial guidance for pelvis trajectory control. OmniControl [23] extends this to any joint at any time. ProgMoGen [11] demonstrates that open-set motion control tasks can be solved by optimizing the diffusion latent code using programmable constraint functions—inspiring our noise optimization approach. DNO [7] further formalizes noise optimization as a universal prior for single-person motion editing, completion, and denoising. EDGE [21] proposes in-betweening for dance, a form of motion inpainting that is related to our long-duration extension.

Multi-Person Interaction Generation. Classical methods synthesize interactions by retargeting or blending captured motion clips [8, 17]. With the rise of learning-based approaches, more flexible interaction generation has become possible. ActFormer [24] uses a GAN-based transformer for multi-person action-conditioned generation. InterGen [10] introduces a two-person diffusion model with weight-sharing denoisers and mutual attention trained on the InterHuman dataset [10]. PriorMDM [15] explores compositional use of single-person priors through a learned communication block (ComMDM) for two-person interaction. InterHandGen [9] addresses two-hand interaction generation using cascaded reverse diffusion. Role-aware interaction generation [19] distinguishes active and passive roles via text conditioning. InterMask [5] uses collaborative masked modeling in discrete space for high-fidelity two-person generation. DiffCollage [27] composes large content via factor graph representations of overlapping diffusion models.

Method	#Person in Network Design	#Person in Training Data	Order of Multi-Person Generation	Method
InterControl [22]	One	One	Sequential	ControlNet
FreeMotion [2]	One	Two	Sequential	ControlNet
Shan et al. [16]	Multiple	Multiple	Concurrent	Supervised Training
ProgMoGen [11]	One	Training-free	N/A (Single-Person)	Noise Optimization
Ours	Two (or multiple)	Training-free	Sequential	Noise Optimization

Table 1. Comparison of recent multi-person interaction generation methods, including single-person method ProgMoGen [11].

Scaling to Arbitrary Groups. Going beyond two persons, InterControl [22] adapts a single-person diffusion model to multi-person scenarios via zero-shot joint-contact guidance. FreeMotion [2] unifies single and multi-person synthesis through decoupled generation and interaction modules but still relies on ControlNet adapters. Shan et al. [16] train a multi-person transformer diffusion model on large-scale datasets curated from internet videos. Table 1 compares these methods along key design dimensions. Unlike prior work, PINO requires no retraining, supports per-pair prompts, and achieves physical plausibility through noise optimization.

3. Preliminaries

Two-Person Diffusion Model. We build on InterGen [10], a denoising diffusion probabilistic model [4] for two-person interaction synthesis. Given a text prompt c, InterGen learns the distribution over two-person motion sequences $\mathbf{x} = (\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) \in \mathbb{R}^{T\times 2\times D}$, where T is the number of frames and $D = 262$ is the per-person feature dimension (encoding joint positions, velocities, rotations, and foot contact). The model uses two weight-sharing transformer denoisers with mutual attention to capture inter-person dependencies. At inference, a DDIM sampler [18] denoises the joint motion from Gaussian noise conditioned on a CLIP [14] text embedding.

Diffusion Noise Optimization. DNO [7] and ProgMoGen [11] demonstrated that, given a pretrained diffusion model ϵ_θ, one can optimize the initial noise $\mathbf{z}_T \sim \mathcal{N}(0, \mathbf{I})$ to satisfy a target objective $\mathcal{L}$ without modifying model weights:

$$\mathbf{z}_T^* = \arg\min_{\mathbf{z}_T} \mathcal{L}(\hat{\mathbf{x}}_0(\mathbf{z}_T)), \tag{1}$$

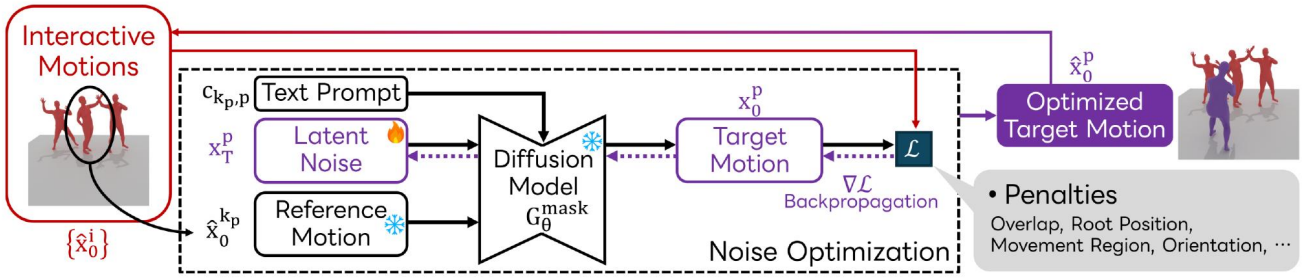

Figure 1. **Person-Interaction Noise Optimization (PINO).** Starting from a two-person generation conditioned on a per-pair prompt, PINO adds characters incrementally. For each new person, person 1 acts as the pivot (reference): its motion is injected into the initial noise via feature masking, and the new person's noise is optimized with physics-based penalties. The same procedure is repeated for long-duration extension via motion inpainting.

where $\hat{\mathbf{x}}_0(\mathbf{z}_T) = \text{Denoise}_\theta(\mathbf{z}_T, c)$ is the sample generated by the full reverse diffusion process from $\mathbf{z}_T$. Gradients flow through the entire denoising chain back to $\mathbf{z}_T$. This enables arbitrary differentiable constraints to be imposed at inference time.

4. Method

4.1. Overview

PINO generates group interactions of arbitrary size in a two-stage pipeline (Fig. 1):

1. **Base generation:** A two-person interaction (persons 1 and 2) is generated with InterGen conditioned on a text prompt c_{12}, with noise optimized to satisfy physics-based penalties.
2. **Sequential extension:** Each additional person k ($k \geq 3$) is generated conditioned on person 1 (the pivot) using a per-pair prompt c_{1k}, with its noise optimized to avoid overlap with all previously generated persons.

For long-duration generation, a third stage extends any two-person sequence using motion inpainting with boundary smoothness penalties.

4.2. Masked Pivot Conditioning

When generating person k ($k \geq 3$) in interaction with the pivot (person 1), we fix person 1's motion $\mathbf{x}^{(1)} \in \mathbb{R}^{T \times D}$ and generate a new pair $(\tilde{\mathbf{x}}^{(1)}, \mathbf{x}^{(k)})$ using InterGen. The key is to constrain $\tilde{\mathbf{x}}^{(1)}$ to match the existing pivot motion. Following the inpainting strategy in the diffusion literature, we construct the initial noise as:

$$\mathbf{z}_T = \mathbf{m} \odot \mathbf{x}^{(1)} + (1 - \mathbf{m}) \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \tag{2}$$

where $\mathbf{m}$ is a binary mask selecting the feature dimensions corresponding to person 1. During each denoising step, the predicted person 1 features are replaced by the fixed pivot motion, ensuring the generated pair remains anchored to the existing group. Person k's noise is left free to be optimized.

4.3. Physics-Aware Noise Optimization

Starting from the masked initial noise, we optimize the free (person k) portion of $\mathbf{z}_T$ by minimizing a composite physics-based loss:

$$\mathcal{L} = \mathcal{L}_{\text{overlap}} + \lambda_r \mathcal{L}_{\text{root}} + \lambda_{\text{reg}} \mathcal{L}_{\text{region}} + \lambda_o \mathcal{L}_{\text{orient}} + \lambda_{\text{rel}} \mathcal{L}_{\text{rel}}, \tag{3}$$

where $\lambda_r, \lambda_{\text{reg}}, \lambda_o, \lambda_{\text{rel}}$ are scalar weights. We describe each penalty below.

Overlap Penalty ($\mathcal{L}_{\text{overlap}}$). Body interpenetration is a common artifact in multi-person generation. We penalize configurations in which any two characters' root (pelvis) positions on the XZ plane are closer than a minimum distance $d_{\min}$ or farther than $d_{\max}$:

$$\mathcal{L}_{\text{overlap}} = \frac{1}{T} \sum_{t=1}^{T} \left[\max(0,\, d_{\min} - \|\mathbf{p}_1^t - \mathbf{p}_k^t\|) + \max(0,\, \|\mathbf{p}_1^t - \mathbf{p}_k^t\| - d_{\text{r}} \right. \tag{4}$$

where $\mathbf{p}_i^t \in \mathbb{R}^2$ is the XZ position of the pelvis of person i at frame t. When $k > 3$, we additionally penalize person k against all previously generated persons (excluding the pivot) using a separate distance threshold d_{other}.

Root Position Penalty ($\mathcal{L}_{\text{root}}$). To control start and/or end positions, we define a target root position $\mathbf{g}_i \in \mathbb{R}^2$ for person i and penalize deviation:

$$\mathcal{L}_{\text{root}} = \max(0, \|\mathbf{p}_i^\tau - \mathbf{g}_i\|^2 - \eta), \tag{5}$$

where τ is the target time frame (start or end) and η is a tolerance margin.

Movement Region Penalty ($\mathcal{L}_{\text{region}}$). When a spatial region constraint is specified (e.g., "person must stay within a 6m $\times$ 6m rectangle centered at the origin"), we penalize

frames where the root position violates the boundary:

$$\mathcal{L}_{\text{region}} = \frac{1}{T}\sum_{t=1}^{T} \mathbb{1}[\mathbf{p}_k^t \notin \mathcal{R}] \cdot \text{dist}(\mathbf{p}_k^t, \mathcal{R}), \tag{6}$$

where $\mathcal{R}$ is a circular or rectangular region and $\text{dist}(\cdot, \mathcal{R})$ is the distance from the boundary.

Facing Orientation Penalty ($\mathcal{L}_{\text{orient}}$). To control relative facing directions between persons, we define a target cosine similarity $\hat{s}$ between the forward-facing unit vectors $\mathbf{n}_1^t$ and $\mathbf{n}_k^t$ derived from each person's hip joints:

$$\mathcal{L}_{\text{orient}} = \frac{1}{T}\sum_{t=1}^{T} \max(0,\, \cos_{\min} - \mathbf{n}_1^t \cdot \mathbf{n}_k^t), \tag{7}$$

where $\cos_{\min}$ is a threshold (e.g., 0.9) above which the constraint is considered satisfied.

Relative Position Penalty ($\mathcal{L}_{\text{rel}}$). To specify the desired relative spatial arrangement between persons (e.g., "person k stands to the left of person 1"), we define a target relative displacement vector $\boldsymbol{\delta}_{1k} \in \mathbb{R}^2$ in the XZ plane and penalize deviation from it:

$$\mathcal{L}_{\text{rel}} = \frac{1}{T}\sum_{t=1}^{T} \max\big(0,\, \|(\mathbf{p}_k^t - \mathbf{p}_1^t) - \boldsymbol{\delta}_{1k}\|^2 - \eta_{\text{rel}}\big), \tag{8}$$

where η_{rel} is a tolerance margin allowing small deviations. This penalty complements the overlap penalty by not only preventing characters from being too close, but actively guiding their arrangement into semantically meaningful configurations.

Optimization. We optimize $\mathbf{z}_T$ using Adam with learning rate 1×10^{-2} and early stopping when $\mathcal{L} < \epsilon_{\text{stop}} = 10^{-3}$, running up to $N_{\text{iter}} = 50$ iterations. Loss weights are fixed to $\lambda_r = 10$, $\lambda_{\text{reg}} = \lambda_o = \lambda_{\text{rel}} = 1$ across all experiments and group sizes. Gradients flow through the entire denoising chain by retaining the computation graph during the reverse diffusion. Gradient norms are clipped to 1.0 before each parameter update to prevent instabilities that arise early in the optimization trajectory, when denoising paths are farthest from the data manifold. The optimization is performed on a single GPU and does not modify any model parameters.

4.4. Long-Duration Generation via Inpainting

For long-duration scenarios, we use motion inpainting to extend a completed short sequence $\mathbf{x}_{[0,T_0]}$ by an additional ΔT frames. The extension shares an overlap window of T_{ov} frames with the existing sequence; the noise for these frames is initialized with the existing motion rather than pure Gaussian noise (analogous to Eq. (2) applied to temporal dimensions). An additional boundary smoothness penalty penalizes discontinuities in joint velocities at the transition frame:

$$\mathcal{L}_{\text{smooth}} = \|\dot{\mathbf{x}}_{T_0}^{(i)} - \dot{\mathbf{x}}_{T_0+1}^{(i)}\|^2, \tag{9}$$

where $\dot{\mathbf{x}}_t^{(i)}$ denotes the joint velocity of person i at frame t. The overlap penalty from Eq. (4) is also applied throughout the extended sequence to maintain physical plausibility. The extended sequence is obtained by concatenating the original and newly generated segments at the overlap boundary.

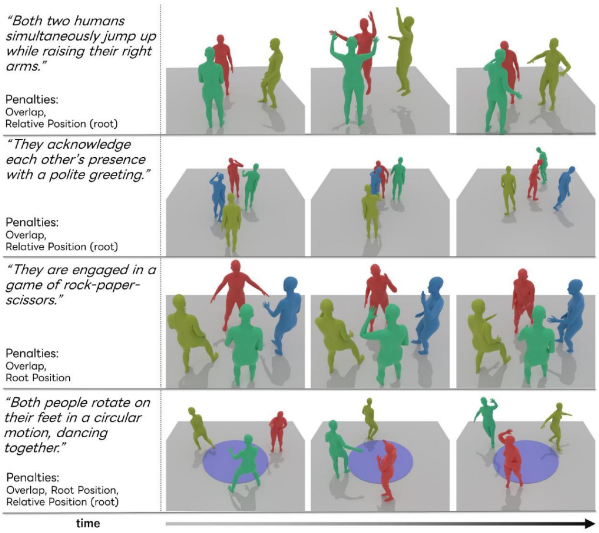

Figure 2. Qualitative results of multi-person interaction generation with PINO. From top to bottom: three-person, four-person, and five-person scenarios. Each group is generated sequentially, with person 1 serving as pivot. Despite increasing group size, characters maintain clear separation and semantically meaningful interactions.

4.5. Per-Pair Prompt Switching

A key advantage of the sequential pairwise generation framework is that each pair $(1, k)$ can be conditioned on a distinct text prompt c_{1k} describing the specific interaction between the pivot and person k. For example, in a five-person group, one might specify:

- c_{12}: "Two people shake hands."
- c_{13}: "Two people high-five."
- c_{14}: "Two people bow to each other."
- c_{15}: "Two people dance together."

This fine-grained control is impossible with shared-prompt methods (e.g., InterGen applied to a single joint prompt for all persons), where all characters tend to perform the same

or similar motion. Our ablation study (Section 5.5) confirms the contribution of each loss component, including how the noise optimization enforces different spatial roles.

5. Experiments

5.1. Experimental Setup

Dataset. We evaluate PINO on the InterHuman dataset [10], a large-scale collection of two-person interaction motion sequences (approximately 107M frames across 23,337 text-annotated interactions). For multi-person evaluation we compose two-person pairs from this dataset, and for five-person evaluation we extend the composition chain.

Baselines. We compare against:

- **InterGen** [10]: The pretrained two-person diffusion model used as our backbone. For multi-person scenarios, InterGen is applied with inpainting (denoted *InterGen (inpainting)*).
- **FreeMotion** [2]: A ControlNet-based approach that supports number-free multi-person generation.

Our method is denoted **PINO-InterGen** to indicate the use of InterGen as the base model.

Metrics. We evaluate along two axes:

- *Non-semantic (physics) metrics*: **Overlap** (fraction of frames with root distance below threshold), **Penetration Volume** (PenVol; cm^3, volumetric interpenetration), **Foot Skate** (foot sliding artifact), and **Max Acceleration** (Max Acc.; physical plausibility of dynamics).
- *Semantic metrics*: **FID** (Fréchet Inception Distance, computed on motion features from a pretrained encoder), **Diversity** (Div.), and **R-Precision** (Top-3; motion-text retrieval accuracy).

Implementation Details. All experiments use InterGen's pretrained checkpoint without any fine-tuning. DDIM sampling runs for $T_{\text{ddim}} = 50$ steps with a classifier-free guidance scale of 2.5. Noise optimization for each additional person requires a separate gradient-enabled forward pass through the full denoising chain. Distance thresholds for the overlap penalty are set to $d_{\min} = 0.5\,\text{m}$ and $d_{\max} = 2.0\,\text{m}$ for pivot pairs, and $d_{\text{other}} = 0.3\,\text{m}$ for non-pivot pairs. For long-duration generation, the temporal overlap window is fixed at $T_{\text{ov}} = 30$ frames. All hyperparameters are held constant across group sizes and datasets, demonstrating the robustness of PINO to hyperparameter choices.

5.2. Two-Person Interaction Generation

Table 2 reports results for two-person interaction generation with overlap avoidance. PINO achieves essentially zero overlap (0.000) and dramatically reduces penetration volume (275.65 cm^3) compared to InterGen (0.119 overlap, 3112.72 cm^3 PenVol). Remarkably, the semantic quality is preserved or improved: FID drops from 13.278 (InterGen) to 13.163 (PINO-InterGen), and Diversity (7.904 vs. 7.793) and R-Precision (0.675 vs. 0.674) are both slightly better. Ground-truth (GT) statistics are provided as upper bounds on quality. Foot Skate and Max Acceleration remain on par with InterGen, showing that noise optimization does not introduce spurious dynamics.

Method	Overlap ↓	PenVol. (cm^3) ↓	Foot Skate ↓	Max Acc. ↓
GT	0.029	471.75	0.099	0.074
InterGen	0.119	3112.72	**0.124**	**0.034**
PINO-InterGen	**0.000**	**275.65**	0.125	**0.034**

(a) Non-semantic metrics.

Method	FID ↓	Div. →	R-Prec. ↑ (Top3)
GT	0.983	7.921	0.715
InterGen	13.278	7.793	0.674
PINO-InterGen	**13.163**	**7.904**	**0.675**

(b) Semantic metrics.

Table 2. Evaluation of two-person interaction generation while avoiding overlap.

Method	Overlap↓	PenVol.↓	Foot Skate↓	Max Acc.↓
InterGen (inpainting)	0.766	25215.78	0.099	**0.030**
PINO-InterGen	**0.000**	**785.01**	**0.091**	0.031

Table 3. Evaluation of multi-person interaction generation

5.3. Multi-Person Interaction Generation

Table 3 compares PINO against InterGen with inpainting for multi-person (three-person and beyond) generation. PINO eliminates overlap entirely (0.000 vs. 0.766) and reduces PenVol by a factor of ~ 32 (785.01 vs. 25215.78), while also improving Foot Skate (0.091 vs. 0.099). Max Acceleration is slightly higher for PINO (0.031 vs. 0.030), but the difference is negligible. These results confirm that the sequential pairwise framework with noise optimization effectively prevents physical artifacts as the group grows.

Scaling to Five Persons. Table 4 reports pairwise FID and overlap for groups of up to five persons (where pair $(1, k)$ reports metrics for persons 1 and k). PINO consistently achieves low overlap across all pairs, reaching only 0.069 for pair $(1, 5)$ compared to 0.977 for InterGen and 0.991 for FreeMotion. FID is also lower for all pairs: e.g., 16.911 vs. 19.501 (InterGen) and 25.671 (FreeMotion) for pair $(1, 5)$. Foot Skate and Max Acceleration are slightly higher than InterGen's in some pairs, a minor trade-off for the dramatic reduction in physical artifacts. Fig. 2 illustrates the diversity of generated group interactions.

Method	Pair	FID↓	Overlap↓	Foot Skate↓	Max Acc.↓
FreeMotion	(1,2)	19.383	0.255	**0.112**	**0.034**
	(1,3)	25.184	0.806	**0.122**	0.038
	(1,4)	25.812	0.964	0.127	0.040
	(1,5)	25.671	0.991	0.130	0.042
InterGen	(1,2)	13.100	0.119	0.126	**0.034**
	(1,3)	19.414	0.799	0.125	**0.036**
	(1,4)	19.386	0.937	**0.124**	**0.038**
	(1,5)	19.501	0.977	**0.124**	**0.039**
PINO-InterGen	(1,2)	**12.920**	**0.000**	0.127	**0.034**
	(1,3)	**17.170**	**0.002**	0.135	0.040
	(1,4)	**16.890**	**0.019**	0.138	0.043
	(1,5)	**16.911**	**0.069**	0.142	0.045

Table 4. Evaluation of multi-person interaction generation. From pair (1,2), interactions are generated incrementally using 1 as a pivot. Overlap is among all individuals generated up to that step.

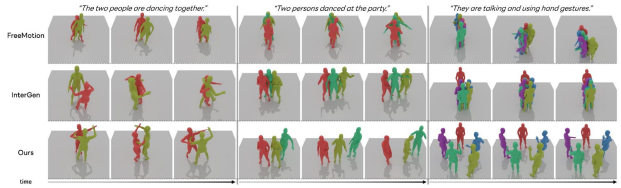

Figure 3. Qualitative comparison of multi-person interaction generation. InterGen (inpainting) produces severe body overlap artifacts. PINO maintains clear separation between all persons across frames.

Method	Overlap↓	PenVol.↓	Foot Skate↓	Max Acc.↓
InterGen (inpainting)	0.672	15622.16	0.070	0.044
PINO-InterGen	**0.031**	**1375.24**	**0.045**	**0.043**

Table 5. Evaluation of non-semantic elements in motion extension.

5.4. Long-Duration Motion Extension

For the motion extension task, we evaluate PINO in two complementary ways.

Non-semantic evaluation. Table 5 shows that the extended motions generated by PINO have significantly lower overlap (0.031 vs. 0.672), lower PenVol (1375.24 vs. 15622.16), and lower Foot Skate (0.045 vs. 0.070) and Max Acceleration (0.043 vs. 0.044) compared to InterGen inpainting. The boundary smoothness penalty effectively prevents discontinuities at the transition frame.

Semantic evaluation. Table 6 evaluates semantic quality by cropping extended sequences to match ground-truth lengths. PINO achieves a lower FID (18.930 vs. 20.390), zero overlap (0.0 vs. 0.089), lower Foot Skate (0.054 vs. 0.081), and lower Max Acceleration (0.025 vs. 0.036). This confirms that noise-optimized inpainting not only avoids artifacts but also maintains the semantic fidelity of the generated motions.

Method	FID↓	Overlap↓	Foot Skate↓	Max Acc.↓
InterGen	20.390	0.089	0.081	0.036
PINO-InterGen	**18.930**	**0.0**	**0.054**	**0.025**

Table 6. Evaluation of semantic elements in two-person motion extension. Extended motions are cropped to match the ground truth length for evaluation.

Method	Pos. Err.↓	Overlap↓	Reg. Viol.↓	Orient. Err.↓
InterGen	1.0	0.292	0.500	1.0
+ $\mathcal{L}_{\text{root}}$	**0.0**	0.333	0.917	1.0
+ $\mathcal{L}_{\text{overlap}}$	**0.0**	**0.0**	0.958	1.0
+ $\mathcal{L}_{\text{region}}$	0.083	**0.0**	**0.043**	1.0
+ $\mathcal{L}_{\text{orientation}}$	0.083	0.043	0.083	**0.208**

Table 7. Ablation study on motion penalties. Metrics include positional errors (Pos. Err.↓), overlap (Overlap↓), region violations (Reg. Viol.↓), and orientation errors (Orient. Err.↓). Positional errors are measured with a threshold of 20 cm, region violations with a threshold of 10 cm, and orientation errors with a threshold of 20 degrees. Lower values indicate better performance.

5.5. Ablation Study

To understand the contribution of each penalty term, Table 7 reports spatio-temporal control metrics as losses are added incrementally. Starting from the vanilla InterGen baseline (all metrics at their worst), adding $\mathcal{L}_{\text{root}}$ reduces positional error to 0.0. Adding $\mathcal{L}_{\text{overlap}}$ eliminates overlap to 0.0. Adding $\mathcal{L}_{\text{region}}$ reduces region violations from 0.958 to 0.043. Finally, adding $\mathcal{L}_{\text{orient}}$ brings orientation error from 1.0 down to 0.208—the residual error reflects the inherent ambiguity of orientation in complex motions. Notably, each penalty targets its corresponding metric without significantly degrading others, validating the modular design of our loss function.

5.6. Runtime Analysis

Noise optimization introduces additional inference time relative to vanilla diffusion sampling. On a single NVIDIA A100 GPU, generating a two-person sequence of $T = 150$ frames with PINO requires approximately 90 seconds, compared to under 5 seconds for standard InterGen. For a five-person group, the total generation time scales approximately linearly with group size, reaching roughly 6 minutes. Although longer than baseline sampling, this overhead avoids any retraining—a process requiring hours on the full InterHuman dataset. The per-person optimization steps are independent given the pivot motion and can therefore be parallelized across GPUs to reduce wall-clock time further.

5.7. Discussion

Taken together, the experimental results reveal a consistent theme: PINO's training-free noise optimization effectively decouples physical plausibility from semantic quality. Unlike approaches that enforce physical constraints through auxiliary loss terms during training—which risk overfitting

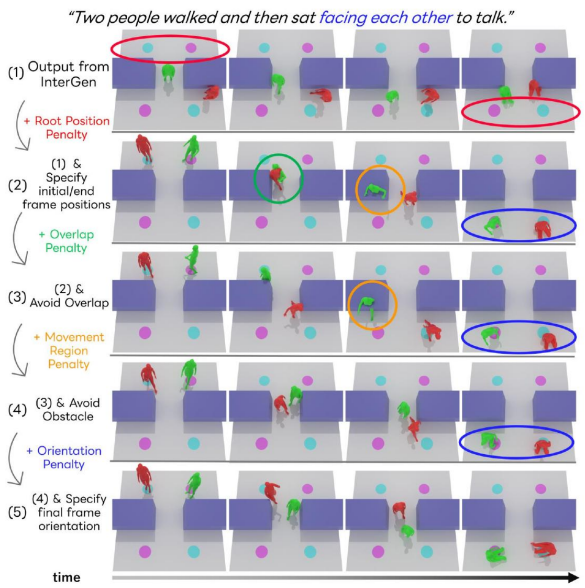

Figure 4. Visualization of the ablation study. Left to right: InterGen baseline, adding $\mathcal{L}_{\text{root}}$, $+\mathcal{L}_{\text{overlap}}$, $+\mathcal{L}_{\text{region}}$, $+\mathcal{L}_{\text{orient}}$. Each penalty progressively enforces its corresponding spatial constraint while preserving overall motion quality.

the model to specific physics regimes—our inference-time optimization adapts to any spatio-temporal constraint expressible as a differentiable penalty. This flexibility makes PINO particularly well-suited for applications requiring on-the-fly customization, such as interactive animation tools or scenario-specific motion synthesis for robotics simulation. Moreover, the ablation results in Section 5.5 confirm that each penalty term targets its respective artifact independently, which simplifies hyperparameter tuning: practitioners can selectively enable or disable individual constraints without destabilizing the overall optimization. The consistent gains in physical plausibility metrics across two-person, multi-person, and long-duration settings further validate the generality of the pivot-anchored pairwise decomposition as a scalable interaction generation paradigm.

One notable observation is that noise optimization acts as a form of post-hoc constraint satisfaction rather than a learned prior; consequently, the quality of generated motions is bounded by the capacity of the underlying InterGen model. When a text prompt specifies an interaction underrepresented in InterGen's training distribution, the optimizer may require additional iterations to balance the semantic and physical objectives simultaneously. A promising direction for future work is to combine our inference-time optimization with lightweight fine-tuning on in-domain data, achieving improved semantic fidelity at lower computational cost. Additionally, the pairwise generation paradigm assumes group interactions can be adequately captured through bilateral relationships with a shared pivot. While this holds for common scenarios—group conversations, coordinated exercises, or team handshakes—it may be less appropriate for interactions requiring simultaneous coupling among three or more persons, such as a group lift, where higher-order dependencies cannot be cleanly decomposed into pairwise terms.

It is also worth highlighting that PINO's modular penalty design provides a natural interface for domain experts to specify interaction semantics. A choreographer, for instance, could define custom region and orientation constraints to stage a group dance performance, while a robotics engineer could specify tight spacing and collision-avoidance distances for coordinated manipulation tasks. This versatility positions noise optimization as a general-purpose inference-time control mechanism for generative motion models, extending well beyond the academic benchmarks considered here.

6. Limitations

While PINO achieves strong results, several limitations remain. First, the overlap penalty operates on root (pelvis) distances and does not model the full 3D body mesh; in highly contorted poses, slight limb penetrations can still occur. Second, noise optimization increases inference time compared to vanilla diffusion sampling, as gradients must be backpropagated through the entire denoising chain at each optimization iteration. Third, as the group size grows, the pivot-anchored approach may limit the diversity of person k's motion, since it is always conditioned on person 1. Exploring alternative graph topologies (e.g., ring or tree structures) for pivot assignment is left as future work. Finally, the current framework generates all persons' motions in a fixed XZ ground plane, which may not capture all interaction geometries (e.g., stacked, airborne motions). Furthermore, backpropagating gradients through the full DDIM denoising chain requires retaining the entire computation graph in GPU memory, which can become a bottleneck for very long sequences ($T \gg 150$) or large groups on memory-constrained hardware; gradient checkpointing strategies could mitigate this cost in future implementations.

7. Conclusion

We have presented PINO, a training-free framework for generating physically plausible multi-person interactions of arbitrary group size. By decomposing group dynamics into a sequence of pairwise diffusion generations anchored by a pivot character, and by optimizing the initial noise with differentiable physics-based penalties, PINO achieves near-zero body overlap and dramatically reduced penetration volumes without any additional training. Per-

pair text prompt switching enables fine-grained role assignment within large groups. A motion inpainting strategy with boundary smoothness penalties extends the framework to long-duration sequences. Extensive experiments on InterHuman demonstrate consistent improvements in physical plausibility metrics while maintaining competitive semantic quality.

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